The empirical recurrence for OEIS A269218, recovered from its tail

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A269218 records "Empirical recurrence of order 52" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 58$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 54$. Its last coefficient is $c_{52} = -12230590464 \neq 0$, so no further tail satisfies anything shorter; any order-52 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A269218 is "Number of 5Xn 0..3 arrays with some element plus some horizontally, diagonally or antidiagonally adjacent neighbor totalling three exactly once.". It has offset 1 and begins

$$0, 103488, 6712704, 474091776, 30541426560, 1910809190712, 116322554833728, 6948456736101264, \ldots$$

The entry states:

> Empirical recurrence of order 52 (see link above)

As of the "Last modified" line on the live entry (Feb 20 2016 11:33:06, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 58$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 58$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 52: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 52.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 116$ exact terms from $n = 2$ on, it returns order 52 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{52} c_i\, a(n-i),$$

c_1	134	c_{14}	10373356396852351	c_{27}	3812663175648698225720	c_{40}	2581904213139
c_2	−5555	c_{15}	−60754086939121226	c_{28}	−7574530653071458330964	c_{41}	−135048465793
c_3	40102	c_{16}	−33051285026349398	c_{29}	−574179066830594819872	c_{42}	−278254917478
c_4	2180813	c_{17}	1199699106842090362	c_{30}	12074743539694574773360	c_{43}	414488563591
c_5	−39931824	c_{18}	−2880844359215306958	c_{31}	−7523516420261269354976	c_{44}	149215946717
c_6	−166790226	c_{19}	−6654302886822206644	c_{32}	−8208890467537441261024	c_{45}	−31186598630
c_7	8421724654	c_{20}	41793577274204606740	c_{33}	11033688262622171759936	c_{46}	−3626839663
c_8	−31263894293	c_{21}	−31262191727385657988	c_{34}	611266657108466569472	c_{47}	9766882383
c_9	−707746782716	c_{22}	−211942668568397286665	c_{35}	−6726538667009839997440	c_{48}	2751298621
c_{10}	6189048380833	c_{23}	502511377399107222724	c_{36}	2174629191883680302528	c_{49}	−118019790
c_{11}	17441027167310	c_{24}	227593302621635128704	c_{37}	1886083319022281623552	c_{50}	447350833
c_{12}	−400445246224257	c_{25}	−2168934637448373353716	c_{38}	−1177651402879996017152	c_{51}	434185961
c_{13}	877859065288196	c_{26}	1802934237742887575116	c_{39}	−184201700555741571072	c_{52}	−1223059

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 54$, and it is the recurrence OEIS A269218 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{52} - c_1 x^{52-1} - \cdots - c_{52}$. Its constant term is $-c_{52} = 12230590464$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 52 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 52, so $\deg q = 52$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 52, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 12 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 52, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -12230590464.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-52 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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